MIL-P-44206A
18 April 1986
SUPERSEDING
MIL-P-44206
15 January 1986

MILITARY SPECIFICATION

# POTATOES AU GRATIN, THERMOSTABILIZED, FOR MEAL, READY-TO-EAT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This document covers thermostabilized potatoes au gratin in flexible packages for use as a component of the Meal, Ready-to-Eat, Individual and Meal, Flight Feeding, Individual.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-44073 - Packaging and Thermoprocessing of Foods in Flexible Pouches

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks, and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces.

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

U.S. Standards of Identity for Pasteurized Processed Cheese

U.S. Standards of Identity for Margarine

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water, WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

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(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, P.O. Box 2109, Novato, CA 94947.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.2).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-flavors, off-odors, and off-colors.

3.2.1 Potatoes, diced, dehydro-frozen. The dehydro-frozen potatoes shall be firm, have a good color and flavor, and be of a white flesh variety suitable for canning. The maximum specific gravity for the potatoes shall be 1.075 with a reducing sugar content of not more than 2.0 percent on a dry weight basis. The potatoes shall be mechanically diced to approximately 3/8 by 3/8 by 1/2 inch dimensions.

3.2.2 Cheese, processed, pasteurized, dehydrated. The dehydrated pasteurized processed cheese shall be derived from fresh pasteurized process American cheese complying with the Standards of Identity for Pasteurized Processed Cheese. The dehydrated cheese shall be produced by spray drying. The spray-dried product shall contain not less than 46 percent fat, not less than 32 percent protein, not more than 4 percent moisture, and not more than 1 percent carbohydrate. The dehydrated cheese shall be a uniform light to medium orange color, be uniform in appearance and composition, and possess a characteristic pasteurized processed cheese flavor and aroma. The product shall be free flowing and free from lumps that do not fall apart under light finger pressure and free from specks or burnt particles.

3.2.3 Margarine. Margarine shall conform to the U.S. Standard of Identity for Margarine and shall possess a sweet, pleasing taste and aroma shall be a medium yellow color with a smooth body and texture free of crumbliness, graininess, grittiness, oiliness, or droplets of free moisture.

3.2.4 Shortening, vegetable, powdered. Powdered vegetable shortening shall consist of components which, when combined, will serve as a product whitening agent. The powdered shortening, when used as an ingredient in this product,



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shall withstand the specified thermoprocessing conditions without browning. A typical analysis of the powdered shortening follows:

| | |
|---|---|
| Fat | 75.00 (± 1.50) percent |
| Protein | 5.50 (± 0.50) percent |
| Carbohydrates | 15.00 (± 1.00) percent |
| Moisture | 2.00 percent, maximum |

The powdered vegetable shortening shall comply with MIL-STD-900.

3.2.5 Salt. Salt shall be noniodized, free-flowing, white, refined sodium chloride, with or without anticaking agents.

3.2.6 Starch, waxy maize, modified. Starch shall be white, odorless, finely pulverized, modified waxy maize starch for use in thermostabilized foods. The modified starch shall demonstrate initial viscosity development in the temperature range of 140°F to 170°F and a typical viscosity (be fully hydrated) at common retort temperatures. The starch shall resist breakdown at low pH, under shear stress, and under conditions of cold storage. The starch shall be bland with essentially no cereal or starch taste and shall comply with MIL-STD-90.

3.2.7 Onion powder. Onion powder shall be Fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.8 Pepper, white, ground. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L. from which the outer covering or the outer and inner coverings have been removed. The ground white pepper shall have a characteristic penetrating odor, a hot biting pungent flavor, and a light color. The white pepper shall contain not less than 1.0 mL of volatile oil per 100 grams of ground white pepper and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 40 sieve.

3.2.9 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this document. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formulation.

3.2.10 Water. Water used for formulation, rehydration, blanching, ice-making, and washing shall conform to the National Interim Primary Drinking Water Regulations.

3.3 Preparation and further processing. Preparation and further processing shall be on a continuous basis.



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3.3.1 Dehydro-frozen potato preparation. Dehydro-frozen potato dices shall be handled in such a manner to prevent discoloration and shall be thawed only to the extent necessary to accommodate the filling operation.

3.3.2 Sauce formulation and preparation. The sauce shall be formulated and prepared as follows:

| Ingredients | Percent by weight |
|---|---|
| Water | 75.11 |
| Cheese, processed, dehydrated | 10.50 |
| Margarine | 5.25 |
| Shortening, vegetable, powdered | 5.00 |
| Starch, waxy maize, modified | 3.00 |
| Salt 1/ | 0.65 |
| Onion powder | 0.42 |
| Pepper, white | 0.07 |

1/ The total amount of salt in the sauce formula shall be adjusted as necessary to produce a product that complies with the finished product salt requirements (see 3.6).

a. A slurry shall be made from the starch and part of the water.

b. The remaining ingredients shall be heated in a steam jacketed kettle to a temperature of 180° to 190°F.

c. The starch slurry shall be added and stirred vigorously and continuously until the sauce reaches a temperature of 180° to 190°F.

d. The volume of the final mixture shall be adjusted with water to compensate for evaporation loss during heating and holding.

e. The sauce shall be maintained in the temperature range of 150° to 180°F and filled within 4 hours after preparation.

3.4 Pouch filling and sealing. Each pouch (see 5.1) shall be filled with product so as to conform to the finished product requirements and to the following requirements:

a. Each pouch shall be filled and sealed in accordance with the pouch filling and sealing requirements specified for class 2 of MIL-P-44073.

b. A two stage filling operation shall be used.

c. Each filled and sealed pouch shall be in the retort process within 1 hour after sealing.

3.5 Pouch thermoprocessing. The filled and sealed pouches shall be thermostabilized by retorting until a sterilization value ($F_o$) of not less than 6



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has been achieved. The thermoprocessing operation shall be in compliance with the thermoprocessing requirements for class 2 of MIL-P-44073.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect, insect parts, hair, wood, paper, paint, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, musty, or moldy.

c. There shall be no color foreign to the product.

d. The average net weight shall be not less than 5.0 ounces.

e. No individual pouch shall contain less than 4.5 ounces of product.

f. The average drained weight shall be not less than 2.25 ounces.

g. No individual pouch shall contain less than 2.0 ounces of drained potato dice.

h. The texture of the potato dice shall not be mushy, hard, fibrous, or tough.

i. The potato dice shall be approximately uniform in size and shape.

j. The sauce consistency shall be smooth, without lumps, and shall not be excessively thick or thin.

k. The salt content of the product in any individual pouch shall be not less than 1.0 percent nor greater than 1.5 percent.

l. The product shall show no evidence of excessive heating (materially darkened or scorched).

3.6.1 Palatability. The finished product shall be equal to or better than the preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods", and the plant sanitation requirements of the appropriate Government inspection agency.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this document.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be evident by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analysis, or other such valid documents acceptable to the inspection agency. In addition, prior to use, each ingredient shall be examined organoleptically, as necessary, to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved products.



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4.5.3 Filled and sealed pouch inspection. Inspection of filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-P-44073.

4.5.4 Net weight inspection. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 4.5 ounces shall be scored as a minor defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 0.1 ounce. In addition, the lot shall be rejected if the sample average net weight is less than 5.0 ounces.

4.5.5 Product inspection. The filled and sealed sample pouches shall be held for a minimum of 72 hours at room temperature (65° to 75°F) after completion of the thermoprocessing operation. The pouches shall be heated in boiling water for 10 minutes, opened, and inspected for the defects indicated in table I. The lot size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects.

TABLE I. Product defects 1/ 2/

| Category | Defect |
|---|---|
| Major | |
| 101 | Drained weight of diced potatoes less than 2.0 ounces. 3/ 4/ |
| 102 | Potato texture mushy, hard, fibrous, or tough. 5/ |
| 103 | Potato dice not approximately uniform in size and shape. 5/ |
| 104 | Sauce consistency lumpy or excessively thick or thin. |
| 105 | Product shows evidence of excessive heating (materially darkened or scorched). |

1/ The presence of foreign material (e.g., glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (e.g., burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.



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2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent.)

3/ To determine drained weight, the free liquid in the pouch shall be poured off and the remaining contents shall be poured into a flat bottom container. A minimum of three times the pouch volume of 190° to 212°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated so as to remove the sauce without breaking the potato dices. The contents shall then be poured into a U.S. Standard 1/4-inch sieve in a manner that will distribute the product over the sieve without breaking the potato dices. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at an approximate 45° angle and allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight shall be reported to the nearest 0.1 ounce.

4/ If the sample average drained weight is less than 2.25 ounces, the lot shall be rejected.

5/ Examination for appearance and texture conformance of the diced potatoes shall be made of representative pieces immediately following determination of drained weight.

4.5.6 Salt content testing. The unopened sample pouches shall be warmed in a water bath to melt fat adhering to the inside of the pouches. The sample pouches shall be opened and the entire contents of each pouch shall be separately blended in a Waring blender or equivalent. The salt content of the product from each pouch in the sample shall be determined in accordance with the Official Methods of Analyses of the Association of Official Analytical Chemists, Chapter: Meat and Meat Products. Test results shall be reported to the nearest 0.1 percent. Any result failing to conform to the salt content requirement in 3.6 shall be classified as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.5.7 Packaging inspection. Inspection of packaging shall be in accordance with the quality assurance provisions of MIL-P-44073.

## 5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the requirements of MIL-P-44073 .



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6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document
b. When a first article is required (see 3.1, 4.4, and 6.2).
c. Provisions for approved preproduction samples (see 3.6.1 and 6.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

Project No. 8940-0549

Review activities:

Army - MD, TS
Navy - MC, MS
DP - SS



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*NOTE*: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-P-44206A | POTATOES AU GRATIN, THERMOSTABILIZED, FOR MEAL, READY-TO-EAT |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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